Dissolving Bench

What is actually floating around in a solution, how much of it there is, and why two clear liquids do not behave the same way in a circuit.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/dissolving-bench/

1 Before you open anything

Answer from what you already know. There is no penalty for being wrong here — the point is to have a committed number on paper before the demonstration settles it.

PREDICTION

Beaker A holds 12.0 g of NaCl stirred into 100.0 g of water at 25 °C. Beaker B is identical except that it holds 24.0 g of NaCl — twice the salt, same water. Beaker B's conductivity is how many times Beaker A's?

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Write one sentence saying what reasoning gave you that number.

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2 Three solutes, same mass

Open the demonstration. The **Solute** panel at the top picks what goes in; the **Amount added** slider sets how much. Every trial here uses 100.0 g of water at 25 °C.

1. Leave the solute on **NaCl** and set **Amount added** to **12.0 g**.
2. Fill in the NaCl row of both tables from the **In the beaker** panel and the **Conductivity tester** panel.
3. Switch the solute to **Sucrose**, then to **Hexane**, leaving the slider at 12.0 g each time, and fill in the remaining rows.

Solute	Dissolved (g)	Moles dissolved (mol)	Solution volume (mL)	Concentration (M)	Dissolved particles (M)
NaCl					
Sucrose					
Hexane					

Solute	Free ions	Bulb	Lamp power	Conductivity (mS/cm)	Resistance (Ω)
NaCl					
Sucrose					
Hexane					

a. With NaCl selected, the **Free ions** tile changes its own label. What does it read instead, and what value does it report?

b. The same 12.0 g dissolved in both the NaCl and the sucrose trials, yet the **Concentration** values differ by about a factor of six. Name the quantity responsible and give both of its values.

c. For NaCl, **Dissolved particles** is exactly twice **Concentration**. For sucrose the two are equal. Write the dissolving equation for each solute and use it to justify the factor.

d. Hexane's **Dissolved** value is not zero. Read it, and read the note underneath about where the rest of the 12.0 g went. State both.

3 Turning the amount up

Go back to **NaCl**. Now change one thing only — the **Amount added** slider — and watch three numbers respond in three different ways.

1. Set **Amount added** to each value in the left column in turn.
2. Record the **Concentration**, **Conductivity** and **Lamp power** readouts each time.
3. For the undissolved column, use the mass the beaker labels **undissolved solid**; write 0 if the note says all of it dissolved.

Amount added (g)	Undissolved solid (g)	Concentration (M)	Conductivity (mS/cm)	Lamp power
2.0				
4.0				
8.0				
12.0				
16.0				
24.0				
36.0				
40.0				

a. Doubling the amount added from 12.0 g to 24.0 g multiplies the concentration by ________ and the conductivity by ________. Work both ratios out from your table.

b. In your own words, state how concentration responds to the amount added, and how conductivity responds. They are not the same rule — say what the difference is.

c. Compare your conductivity ratio with the prediction you wrote in Part 1. If they disagree, name the specific assumption in your reasoning that the data does not support.

d. Between 36.0 g and 40.0 g, not one of the three readouts moves. Explain what the solution is doing during that stretch, at the particle level, that keeps every number fixed.

> Lamp power behaves differently again: it climbs steeply at low amounts and then barely moves. The panel under the tester tells you why the circuit is built that way — the solution sits in series with a 100 Ω lamp, so once the solution's own resistance falls well below 100 Ω, the lamp is what limits the current, not the salt.

4 Where the usual shortcut breaks

A very common move is to treat 100.0 g of water as 100.0 mL of solution and divide the moles by 0.1000 L. It is a reasonable first instinct, and it is close enough to pass unnoticed in dilute work. Here is where it stops being close enough.

1. Set up each row below, read **Moles dissolved**, **Solution volume** and **Concentration** from the screen.
2. In the fourth column, compute moles ÷ 0.1000 L yourself.
3. In the last column, give the percent by which your shortcut answer exceeds the demonstration's concentration.

Setting	Moles dissolved (mol)	Solution volume (mL)	Concentration readout (M)	mol ÷ 0.1000 L (M)	% too high
NaCl, 36.0 g					
Sucrose, 24.0 g					
Sucrose, 40.0 g					

a. The shortcut is always too high, never too low. Explain why the error only ever goes in that one direction.

b. The error is much worse for sucrose at 40.0 g than for NaCl at 36.0 g, even though more grams of NaCl went in. Using the **Solution volume** values, say what makes the sucrose case worse.

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c. Now work backwards. With NaCl selected, find the smallest **Amount added** at which undissolved solid first appears on the bottom of the beaker. Record that setting, and state the solubility limit the note quotes.

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d. Your answer to *c* is a slider setting, not a measurement. Explain what the slider's step size means for how precisely this method can pin the solubility down.

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5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it and check yourself — and where you disagree with the screen, find the step that caused it. Use $M(\mathrm{NaCl})$ = 58.44 g/mol, $M(C_{12}H_{22}O_{11})$ = 342.30 g/mol, and 25 °C throughout.

1. 8.0 g of NaCl is stirred into 100.0 g of water. (i) Find the moles dissolved. (ii) Find the molarity assuming the solution volume is 100.0 mL. (iii) The solution volume is really 102.6 mL. Recompute the molarity, and state whether part (ii) was too high or too low.

2. Beaker P holds 24.0 g of NaCl in 100.0 g of water and has a solution volume of 108.6 mL. Beaker Q holds 24.0 g of sucrose in 100.0 g of water and has a solution volume of 115.3 mL. Find the concentration of *dissolved particles* in each, and state how many times more particles per litre Beaker P holds. Give that factor to two significant figures.

3. 40.0 g of NaCl is added to one beaker of 100.0 g of water, and 40.0 g of sucrose to another. NaCl dissolves to 36.0 g per 100 g of water; sucrose to about 211 g per 100 g of water. State how many grams remain undissolved in each beaker. Then explain why the beaker in which everything dissolved is the one that leaves the bulb dark.

4. A lab group reports that their sugar solution made the bulb glow faintly, and concludes that sucrose must ionise a little. Give a better explanation for the glow, and state one measurement they could take that would distinguish their explanation from yours.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Switch the view to **Close-up** for each of the three solutes. For NaCl, note which end of the water molecule turns toward Na^+ and which turns toward Cl^-. Then look at the sucrose close-up: the water molecules there are not all pointed the same way. Work out why not.
- Set the slider to **0 g** and read **Conductivity** and **Resistance**. Neither is zero. Decide what is carrying that current, and whether it should count as an error in the model.
- With hexane selected, set **20.0 g** and read the volume printed on the floating layer. Check it against the density given in the solute blurb, 0.655 g/mL. Does the layer volume account for the whole 20.0 g?
- Uncheck **Show the water shells** and watch the beaker view. Decide what that checkbox is hiding, and whether anything it hides is real chemistry or just drawing. The panel headed *Where the model simplifies* is worth reading before you answer.